

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250287073

Kind Code

A1

Publication Date

September 11, 2025

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REAL-TIME STREAMING AND PLAYBACK OF DIGITAL MEDIA

Abstract

The present technology provides systems and methods to facilitate seek functionality in real-time streaming and playback of digital media. In particular, the present technology provides for generating an index for a digital media file encoded in a non-streaming format, determining a portion of the digital media file to transcode, and transcoding the portion of the digital media file to a proxy format based on the index.

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Appl. No.: 19/218072

Filed: May 23, 2025

Related U.S. Application Data

parent US continuation 18451491 20230817 parent-grant-document US 12335580 child US 19218072

Publication Classification

Int. Cl.: H04N21/6587 (20110101); G11B27/10 (20060101); H04N21/2343 (20110101); H04N21/239 (20110101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/451,491, filed Aug. 17, 2023, entitled “REAL-TIME STREAMING AND PLAYBACK OF DIGITAL MEDIA”, which is hereby incorporated herein by reference in its entirety.

FIELD OF INVENTION

[0002] The present technology relates to digital media. More particularly, the present technology relates to real-time streaming, including efficient seek operations, of digital media encoded in formats that do not natively support such operations.

BACKGROUND

[0003] Advances in digital media technologies have introduced various digital media formats. For example, MP4 is a commonly used digital multimedia container format that stores video and audio. The International Organization for Standardization/International Electrotechnical Commission (ISO/IEC) standardized the MP4 format with the intention of providing a general media file format for use with web applications and other online services. Other digital media formats are designed for different purposes. For example, the MPEG Transport Stream format, independent of the underlying codec, does not contain a seekable index (e.g., an index that supports seek functions). Such digital media formats do not natively support real-time streaming with seek functionality over the Internet. For these and other reasons, various digital media technologies suffer from various drawbacks and inefficiencies.

SUMMARY

[0004] Various aspects of the present technology relate to systems and methods to facilitate seek functionality in real-time streaming and playback of digital media. The present technology provides for generating an index for a digital media file encoded in a non-seekable format, determining a portion of the digital media file to transcode, and transcoding the portion of the digital media file to a proxy format based on the index.

[0005] According to another aspect of the present technology, generating the index for the digital media file comprises: performing a pass through the digital media file, and identifying during the pass through the digital media file, file offsets of key frames of the digital media file based on frame headers.

[0006] According to another aspect of the present technology, generating the index for the digital media file comprises: storing a file offset and a time offset for each key frame of the digital media file, wherein each file offset provides a respective location within the digital media file for each key frame, and wherein each time offset provides a respective time associated with each key frame.

[0007] According to another aspect of the present technology, determining the portion of the digital media file to transcode is based on a seek operation, and transcoding the portion of the digital media file comprises: determining a file offset corresponding to a time in the digital media file indicated by the seek operation based on the index, and transcoding the digital media file from the file offset.

[0008] According to another aspect of the present technology, determining the portion of the digital media file to transcode is based on a clip operation, and transcoding the portion of the digital media file comprises: determining a starting file offset corresponding to a start time in the digital media file indicated by the clip operation based on the index, determining an ending file offset corresponding to an end time in the digital media file indicated by the clip operation based on the

index, and transcoding the digital media file from the starting file offset to the ending file offset.

[0009] According to another aspect of the present technology, determining the portion of the digital media file to transcode is based on a playback operation, and transcoding the portion of the digital media file comprises: determining a file offset corresponding to a beginning of the portion of the digital media file based on the index, and transcoding the portion of the digital media file from the file offset.

[0010] According to another aspect of the present technology, the present technology further provides for storing the transcoded portion of the digital media file, and providing the transcoded portion of the digital media file in response to a user request.

[0011] According to another aspect of the present technology, the present technology further provides for determining a first vector embedding based on a frame in the digital media file, determining a second vector embedding based on a search query, wherein determining the portion of the digital media file to transcode is based on the first vector embedding and the second vector embedding, and providing the transcoded portion of the digital media file in response to the search query.

[0012] According to another aspect of the present technology, the portion of the digital media file is transcoded without transcoding other portions of the digital media file.

[0013] According to another aspect of the present technology, the proxy format is a streaming format.

[0014] These and other objects, features, and characteristics of the present technology, as well as the methods of operation and functions of the related elements of structure and the combination of parts and economies of manufacture, will become more apparent upon consideration of the following description and the appended claims with reference to the accompanying drawings, all of which form a part of this specification, wherein like reference numerals designate corresponding parts in the various figures. It is to be expressly understood, however, that the drawings are for the purpose of illustration and description only and are not intended as a definition of the limits of the invention. As used in the specification and in the claims, the singular form of “a”, “an”, and “the” include plural referents unless the context clearly dictates otherwise.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] FIG. 1 illustrates an example system configured to facilitate real-time streaming and playback of digital media, in accordance with one or more implementations.

[0016] FIGS. 2A-2C illustrate example user interfaces related to real-time streaming and playback of digital media, in accordance with one or more implementations.

[0017] FIGS. 3A-3B illustrate example user interfaces related to real-time streaming and playback of digital media, in accordance with one or more implementations.

[0018] FIG. 4 illustrates an example method related to real-time streaming and playback of digital media, in accordance with one or more implementations.

[0019] The figures depict various objects, features, and characteristics of the present technology for purposes of illustration. One skilled in the art will readily recognize from the following discussion that alternative structures and methods illustrated in the figures can be employed without departing from the principles of the present technology described herein.

DETAILED DESCRIPTION

[0020] The present technology relates to real-time streaming and playback of digital media. In general, various digital media formats do not natively support seek functionality in real-time streaming and playback. With such digital media formats, users generally cannot navigate (e.g., seek, clip) to a particular location in a file and must begin playback from the beginning of the file.

One approach to providing the navigation operations to the file is to create a new proxy version of the file in a different format. This proxy version is typically created before the users try to access the file so that the proxy version is ready for streaming and playback. However, this approach is highly costly with regard to storage and computational effort as this approach requires duplication of files in different formats. Accordingly, the present technology addresses these and other drawbacks by providing various features for seek functionality in real-time streaming and playback of digital media files.

[0021] To facilitate these and other features, the present technology provides for preprocessing a media file to extract frame offset data (e.g., byte offset) from the media file for key frames in the media file. The frame offset data can be stored as index entries in an index that maps time offsets of the key frames to the frame offset data, or byte offset in the media file. The index can be stored in a database or as a separate file. As this preprocessing does not require decoding, encoding, or transcoding the media file, this preprocessing is less resource-intensive, and therefore less costly, than converting the media file into a proxy version. To facilitate streaming of the media file, the index can be retrieved and provided for efficient seek operations, clip operations, and playback operations. The media file can be transcoded in real-time to provide streaming and playback of the media file at particular locations in the media file in response to the seek operations, clip operations, and playback operations. Any transcoded portions of the media file can be stored and served again as appropriate. In this way, the cost of decoding, encoding, and transcoding is deferred until playback and applied to portions of the media file that are requested by the users. In various applications, such as media post-processing, only a portion of a media file may be played back. Therefore, the present technology provides for improvements in functionality and cost efficiency for real-time streaming and playback of digital media files.

[0022] FIG. 1 illustrates an example system **100** configured to facilitate seek functionality in real-time streaming and playback of digital media, according to one or more implementation of the present technology. System **100** may include, without limitation, client computer systems **110a**, **110b**, a computer system **130**, and/or other components. Various functionality described herein with respect to system **100** may be described as being performed by the client computer systems **110a**, **110b**, the computer system **130**, and/or other components. In other implementations, some or all of the functionality described herein may be performed by any of the client computer systems **110a**, **110b**, the computer system **130**, and/or other components.

[0023] In FIG. 1, in some implementations, client computer systems **110a**, **110b**, a computer system **130**, and/or other components may be operatively linked via one or more electronic communication links. For example, such electronic communication links may be established, at least in part, via a network **102** such as the Internet, Bluetooth, and/or other networks. It will be appreciated that this is not intended to be limiting, and that the scope of this disclosure includes implementations in which client computer systems **110a**, **110b**, a computer system **130**, and/or other components may be operatively linked via some other communication media.

[0024] Client computer systems **110a**, **110b** may include one or more processors **112a**, **112b**, one or more storage devices **114a**, **114b**, which can store playback modules **116a**, **116b**, and/or other components. Processors **112a**, **112b** may be programmed to perform various operations based on one or more computer program instructions. For example, processors **112a**, **112b** may be programmed to perform various operations by the playback module **116a**, **116b** and/or other instructions. Illustration of one or more processors **112a**, **112b** in FIG. 1 is not intended to be limiting. The one or more processors **112a**, **112b** may include a plurality of hardware, software, and/or firmware components operating together to provide the functionality attributed herein to one or more processors **112a**, **112b**. For example, one or more processors **112a**, **112b** may be implemented by a cloud of computing platforms operating together as one or more processors **112a**, **112b**.

[0025] One or more storage devices **114a**, **114b** may comprise non-transitory storage media that

electronically stores information. The electronic storage media of one or more storage devices **114a**, **114b** may include one or both of system storage that is provided integrally (i.e., substantially non-removable) with one or more processors **112a**, **112b** and/or removable storage that is removably connectable to one or more processors **112a**, **112b** via, for example, a port (e.g., a USB port, a firewire port, etc.) or a drive (e.g., a disk drive, etc.). One or more storage devices **114a**, **114b** may include one or more of optically readable storage media (e.g., optical disks, etc.), magnetically readable storage media (e.g., magnetic tape, magnetic hard drive, floppy drive, etc.), electrical charge-based storage media (e.g., EEPROM, RAM, etc.), solid-state storage media (e.g., flash drive, etc.), and/or other electronically readable storage media. One or more storage devices **114a**, **114b** may include one or more virtual storage resources (e.g., cloud storage, a virtual private network, and/or other virtual storage resources). One or more storage devices **114a**, **114b** may store software algorithms, information determined by one or more processors **112a**, **112b**, information received from computer system **130**, and/or other information that enables client computer system **110a**, **110b** to function as described herein.

[0026] Playback module **116a**, **116b** can facilitate upload of media data, download (e.g., streaming) of media data, and playback of media data. The upload, download, and playback of the media data can be facilitated by an application, such as a web browser or a media streaming application. The media data may include, for example, video data and/or video files. The media data may be encoded with one of various codecs and stored in one of various file formats. For example, media data can be stored in an MPEG transport stream file, which does not contain a frame index. The client computer system **110a** can upload the media data to computer system **130** through a media application on the client computer system **110a**. The media data can be stored on the computer system **130**, for example, in data store **140**. Client computer system **110b** can stream the media data uploaded by computer system **110a** and playback the media data through a web browser.

[0027] Computer system **130** can include one or more processors **132**, one or more storage devices **134**, which can store a preprocessing module **136**, a transcoding module **138**, a search module **139**, and/or other instructions, and/or other components. Processors **132** may be programmed to perform various operations based on one or more computer program instructions. For example, processors **132** may be programmed to perform various operations by the preprocessing module **136**, the transcoding module **138**, the search module **139**, and/or other instructions. Illustration of one or more processors **132** in FIG. **1** is not intended to be limiting. The one or more processors **132** may include a plurality of hardware, software, and/or firmware components operating together to provide the functionality attributed herein to one or more processors **132**. For example, one or more processors **132** may be implemented by a cloud of computing platforms operating together as one or more processors **132**.

[0028] One or more processors **132** may be configured to provide information processing capabilities in computer system **130**. As such, one or more processors **132** may include one or more of a digital processor, an analog processor, a digital circuit designed to process information, an analog circuit designed to process information, a state machine, and/or other mechanisms for electronically processing information. Although one or more processors **132** is shown in FIG. **1** as a single entity, this is for illustrative purposes only. In some implementations, one or more processors **132** may include a plurality of processing units. These processing units may be physically located within the same device, or one or more processors **132** may represent processing functionality of a plurality of devices operating in coordination. One or more processors **132** may be configured to execute instructions provided by preprocessing module **136**, transcoding module **138**, search module **139**, and/or other components. One or more processors **132** may be configured to execute instructions provided by preprocessing module **136**, transcoding module **138**, search module **139**, and/or other components by software; hardware; firmware; some combination of software, hardware, and/or firmware; and/or other mechanisms for configuring processing capabilities on one or more processors **132**. As used herein, the term “component” may refer to any component or set

of components that perform the functionality attributed to the component. This may include one or more physical processors during execution of processor readable instructions, the processor readable instructions, circuitry, hardware, storage media, or any other components.

[0029] It should be appreciated that although preprocessing module **136**, transcoding module **138**, and search module **139** are illustrated in FIG. 1 as being implemented within a single computer system **130**, one or more of preprocessing module **136**, transcoding module **138**, and search module **139** may be implemented remotely. The description of the functionality provided by these modules is for illustrative purposes, and is not intended to be limiting, as any of preprocessing module **136**, transcoding module **138**, and search module **139** may provide more or less functionality than is described. For example, one or more of preprocessing module **136**, transcoding module **138**, and search module **139** may be eliminated, and some or all of its functionality may be provided by other ones of preprocessing module **136**, transcoding module **138**, and search module **139**. As another example, one or more processors **132** may be configured to execute one or more additional components that may perform some or all of the functionality attributed below to one of preprocessing module **136**, transcoding module **138**, and search module **139**.

[0030] One or more storage devices **134** may comprise non-transitory computer-readable storage media that electronically stores information. The non-transitory computer-readable storage media of the one or more storage devices **134** may include one or both of system storage that is provided integrally (i.e., substantially non-removable) with one or more processors **132** and/or removable storage that is removably connectable to one or more processors **132** via, for example, a port (e.g., a USB port, a firewire port, etc.) or a drive (e.g., a disk drive, etc.). One or more storage devices **134** may include one or more of optically readable storage media (e.g., optical disks, etc.), magnetically readable storage media (e.g., magnetic tape, magnetic hard drive, floppy drive, etc.), electrical charge-based storage media (e.g., EEPROM, RAM, etc.), solid-state storage media (e.g., flash drive, etc.), and/or other electronically readable storage media. One or more storage devices **134** may include one or more virtual storage resources (e.g., cloud storage, a virtual private network, and/or other virtual storage resources). One or more storage devices **134** may store software algorithms, information determined by one or more processors **132**, information received from client computer system **110a**, **110b**, and/or other information that enables computer system **130** to function as described herein.

[0031] Preprocessing module **136** can facilitate preprocessing digital media to extract frame offset data from the digital media. Preprocessing the digital media can involve a pass (e.g., single pass) through a digital media file. During the pass, the digital media file is analyzed to identify frames without decoding the digital media in the digital media file. The frames can be identified based on frame headers, or other reference points, within the digital media file. For each identified frame, a file offset can be stored in an index (or index file) associated with the digital media file. The file offset can indicate where in the digital media file the identified frame is located. For example, the file offset can provide a number of bytes from the beginning of the digital media file to where the identified frame is located. For each identified frame, a time offset can be stored in the index associated with the digital media file. The time offset can indicate a time associated with when the identified frame is to be played back. For example, the time offset can provide a number of seconds from the beginning of the digital media to when the identified frame is to be played back. In some cases, a digital media file can include multiple tracks of digital media. In these cases, track information associated with each identified frame can be stored in the index to maintain the different tracks of digital media. The generated index can facilitate efficient seek operations, clip operations, and playback operations for the digital media file by providing a mapping of file offsets and time offsets for frames in the digital media file.

[0032] In some implementations, preprocessing the digital media file can involve extracting metadata from the digital media file. The metadata can include information associated with the digital media file, such as an author, a title, and a location associated with the digital media file.

The metadata can facilitate search functionality for the digital media file. For example, a search for a particular author can return digital media files with metadata that include the particular author. In some implementations, preprocessing the digital media file can involve extracting thumbnail images from the digital media file. Extracting the thumbnail images can be performed in a second pass through the digital media file subsequent to a first pass to extract frame offset data from the digital media file. During the pass, frames in the digital media file can be sampled and thumbnail images can be extracted from the sampled frames. The extracted thumbnail images can be provided as previews when navigating to a particular location in the digital media file to facilitate efficient seeking or clipping during playback. For example, during a pass of a digital media file, frames in the digital media file can be sampled so that 50 thumbnail images are extracted during the pass. When the digital media file is provided for playback and a seek operation is performed on the digital media file, the 50 thumbnail images can provide previews of what the sampled frames display. In this way, a user is provided with a preview of various points in the digital media file, which helps to facilitate an efficient seek operation, clip operation, or playback operation.

[0033] As an example of the above, a video uploaded to the computer system **130** can be preprocessed to generate an index file for the video. In a pass through the video, video frame headers can be identified for the video. The video frames in the video can be identified using the video frame headers. For each video frame that is identified, a file offset and a time offset is stored in the index file. For example, the first frame of the video can have an entry corresponding to where the frame is in the file (e.g., X bytes from the beginning of the video file) and when the frame is played back during playback of the video (e.g., 0 seconds from the beginning of the video). In this way, the index file generated for the digital video file facilitates efficient seeking, clipping, and playback of the digital video file. For example, a user can begin playback of the video at a selected time (e.g., 3 minutes from the beginning of the digital video) and, based on the mapping provided by the index file, playback of the digital video can begin at a file offset corresponding with the selected time. Furthermore, the index file can be generated with more computational efficiency than transcoding the entire digital video file into a proxy format that supports seek operations and clip operations.

[0034] Transcoding module **138** can facilitate transcoding of digital media files. The transcoding module **138** can transcode a portion of the digital media file in response to a seek operation, a clip operation, or a playback operation. The portion of the digital media file that is transcoded can be determined based on an index associated with the digital media file. In response to a seek operation, the index associated with the digital media file can be used to identify a file offset corresponding to a time in the digital media file indicated by the seek operation. The digital media file can be transcoded from the identified file offset and provided for playback until playback of the digital media file stops (e.g., by a stop operation or by reaching the end of the digital media file). For example, a user can seek to a particular time (e.g., 3 minutes from the beginning) in a digital media file. Using the index associated with the digital media file, a file offset mapped to the particular time can be determined. The digital media file can be transcoded from the file offset and played back for the user. In response to a clip operation, the index associated with the digital media file can be used to identify a starting file offset corresponding with a start time of the clip and an ending file offset corresponding with an end time of the clip. The digital media file can be transcoded from the starting file offset to the ending file offset, generating a clip of the digital media file that has been transcoded (e.g., to a streaming format). For example, a user can clip a digital media file from a start time (e.g., 3 minutes from the beginning) to an end time (e.g., 5 minutes from the beginning). Using the index associated with the digital media file, a starting file offset mapped to the start time of the clip and an ending file offset mapped to the end time of the clip can be determined. The digital media file can be transcoded from the starting file offset to the ending file offset to generate a transcoded clip of the digital media file. In response to a playback operation, the digital media file can be transcoded from the beginning of the digital media file until

the playback operation is stopped. The portions of the digital media file generated in these ways can be stored for future use. That is, in response to a seek operation, clip operation, or playback operation that involves a portion of a digital media file that was previously transcoded, the previously transcoded portion of the digital media file can be served again.

[0035] As an example of the above, a user can select a video to stream from the computer system **130**. The video can be stored in a non-streaming format. The video can be transcoded starting from the beginning of the video and served to the user. To do so, an index file associated with the video can be used to determine a file offset of the first frame of the video. Transcoding and playback of the video can begin from the file offset corresponding to the first frame of the video. During playback of the video, the user can skip, or seek, to another point in the video (e.g., 30 minutes ahead into the video). The video can be transcoded starting from the seek point by using the index file to determine a file offset corresponding with the seek point. Transcoding and playback of the video can begin from the file offset corresponding to the seek point. When the user stops playback of the video, two transcoded portions of the video are generated and stored. The first portion can include transcoded video from the beginning of the video to where playback was stopped when the user performed the seek operation. The second portion can include transcoded video from the seek point to where playback was stopped by the user. The first portion and the second portion can be served to the user, or subsequent users, in response to requests that involves those portions of the video. As illustrated herein, the present technology serves media requested by a user without transcoding an entire digital media file, thereby improving computational efficiency and storage costs. These improvements are especially advantageous in situations where only a small portion of video material in a system is played over the lifetime of the system or when costs related to the system are based on use of the system rather than number of files managed within the system.

[0036] Search module **139** can facilitate searching of digital media files. In some implementations, searching digital media files can involve a text search for metadata extracted from the digital media files. During preprocessing of the digital media files, metadata containing information such as author, title, and location associated with the digital media files can be extracted. The search module **139** can receive a search query and provide digital media files with information matching the search query. For example, a user can search for videos by a particular author by entering a name of the particular author as a search query. Videos with metadata containing information that matches the name entered as the search query can be provided to the user as search results.

[0037] In some implementations, search module **139** can facilitate searching of digital media files based on semantic vector embeddings. In general, vector embeddings can be numerical representations (e.g., vectors) of various objects and concepts, such as those represented in images and text. Vector embeddings can be mapped to an embedding space and compared to determine relationships between the vector embeddings. The search module **139** can generate semantic vector embeddings for images (e.g., thumbnail images, key frames, frames) extracted from a digital media file. The images can be extracted during preprocessing of the digital media file. During preprocessing, frames in the digital media file can be sampled. The frames can be sampled, for example, at a predetermined rate (e.g., every 20 frames, every key frame), to meet a predetermined threshold (e.g., 50 frames, 50 key frames), or based on detected scene changes. The scene changes can be detected, for example, based on changes in pixel values (e.g., DCT DC coefficients) between frames that exceed a threshold pixel value. Images can be extracted from the sampled frames. Semantic vector embeddings can be generated based on the images. The semantic vector embeddings can represent objects or concepts depicted in the images. The semantic vector embeddings can be mapped to a vector embedding space with vector embeddings generated based on search queries. The semantic vector embeddings of the images can be evaluated with the vector embeddings of the search queries based on a cosine similarity. The semantic vector embeddings of the images that satisfy a threshold cosine similarity with a vector embedding of a search query can be determined to satisfy the search query. Clips or playback positions of the digital media file that

correspond with the images of the semantic vector embeddings that satisfy the search query can be provided in response to the search query. For example, a video can be sampled at detected scene changes, and images can be extracted from the frames at the detected scene changes. Semantic vector embeddings can be generated based on the images. The semantic vector embeddings can be associated in an index file with file offsets and time offsets of the frames from which the semantic vector embeddings were generated. A user can search for videos depicting a particular object by entering words describing the object as a search query. A vector embedding can be generated based on the search query. The semantic vector embeddings can be evaluated with the vector embedding to determine which of the semantic vector embeddings satisfy the search query. Playback positions of the video associated with the semantic vector embeddings that satisfy the search query can be provided as search results.

[0038] Generation of semantic vector embeddings can utilize machine learning methodologies. For example, a machine learning model can be trained to generate semantic vector embeddings based on images extracted from frames of a digital media file. The machine learning model can be trained based on training data that includes instances of labelled images. An instance of training data can include an image and a corresponding label describing contents of the image. As part of the training of the machine learning model, a vector embedding can be generated based on the image. The vector embedding can be mapped to a vector embedding space with other vector embeddings. The machine learning model can be trained so that vector embeddings of images with the same or similar labels are mapped closer together than vector embeddings of images with dissimilar labels. Likewise, a machine learning model can be trained to generate vector embeddings based on text from search queries. The machine learning model can be trained based on training data that includes instances of search queries. As part of the training of the machine learning model, vector embeddings can be generated based on the text from the search queries. The machine learning model can be trained so that vector embeddings of text with the same or similar meanings are mapped closer together than vector embeddings of text with dissimilar meanings. These machine learning models can be trained to use the same vector embedding space so that vector embeddings of images and vector embeddings of text that represent the same or similar concepts are mapped closer together than vector embeddings of images and vector embeddings of text that represent dissimilar concepts.

[0039] FIGS. 2A-2C illustrate example interfaces associated with seek operations, clip operations, and playback operations, according to some implementations of the present technology. The various functionality described herein can be performed by, for example, the system **100** of FIG. 1. It should be understood that there can be additional, fewer, or alternative functionality or steps performed in similar or alternative orders, or in parallel, based on the various features and embodiments discussed herein unless otherwise stated.

[0040] FIG. 2A illustrates an example interface **200** that can facilitate seek operations, clip operations, and playback operations. The example interface **200** displays videos **202a**, **202b**, **202c**, **202d**, **202e**, **202f** available for viewing. The videos **202a**, **202b**, **202c**, **202d**, **202e**, **202f** can include videos that are encoded in a non-streaming format (e.g., MPEG transport stream, Digital Nonlinear Extensible High Resolution/Definition (DNxHR/DNxHD)). As illustrated in FIG. 2A, video **202b** (“Movie_002”) is selected, and the example interface provides various options for operations available for video **202b**. The example interface **200** includes a clip option **204** to perform a clip operation to generate a clip based on the selected video **202b**. The example interface **200** includes a progress bar **212** and a seek pointer **208** to facilitate seek operations for the selected video **202b**. For example, a user can drag the seek pointer **208** along the progress bar **212** to seek to a particular point in the selected video **202b**. The example interface **200** includes a play button **210** to facilitate playback operations for the selected video **202b**. For example, a user can select the play button **210** to begin playback of the selected video **202b** from a particular point indicated by the seek pointer **208**. In response to selection of the play button **210**, the selected video **202b** can be transcoded and

provided for display in video window **206**.

[0041] FIG. 2B illustrates an example interface **230** that can facilitate clip operations. The example interface **230** can, for example, be provided in response to a user selection of the clip option **204** in FIG. 2A. As illustrated in FIG. 2B, the example interface **230** includes a save clip option **232** to facilitate saving of a clip. For example, a user can select the save clip option **232** to generate a clip in a proxy format. The example interface **230** includes a preview bar **238** that includes thumbnail images of sampled frames. The example interface **230** includes a starting time indicator **234** and an ending time indicator **236** for selecting the start and the end of the clip. For example, a user can slide the starting time indicator **234** to a desired starting time for the clip and slide the ending time indicator **236** to a desired ending time for the clip. Selection of the save clip option **232** can generate a clip from the starting time indicated by the starting time indicator **234** to the ending time indicated by the ending time indicator **236**.

[0042] FIG. 2C illustrates an example interface **260** that can facilitate clip operations. The example interface **260** can, for example, be provided in response to user interaction with the starting time indicator **234** and the ending time indicator **236** of FIG. 2B. As illustrated in FIG. 2C, the example interface **260** includes a starting time indicator **264** and an ending time indicator **266** for selecting the start and the end of the clip. As illustrated in this example, a user has slid the starting time indicator **264** from the start of the video to a desired starting time for the clip. Likewise, the user has slid the ending time indicator **236** from the end of the video to a desired ending time for the clip. Selection of the save clip option **262** can generate a clip from the starting time indicated by the starting time indicator **264** to the ending time indicated by the ending time indicator **266**.

[0043] FIGS. 3A-3B illustrate example interfaces associated with search operations, according to some implementations of the present technology. The various functionality described herein can be performed by, for example, the system **100** of FIG. 1. It should be understood that there can be additional, fewer, or alternative functionality or steps performed in similar or alternative orders, or in parallel, based on the various features and embodiments discussed herein unless otherwise stated.

[0044] FIG. 3A illustrates an example interface **300** that can facilitate search operations. As illustrated in FIG. 3A, the example interface **300** displays videos **302a**, **302b**, **302c**, **302d**, **302e**, **302f**, **302g**, **302h**, **302i**, **302j** available for viewing. The example interface **300** provides a search bar **304** to facilitate search operations. For example, a user can enter a search query in the search bar **304** to perform a search operation based on the search query.

[0045] FIG. 3B illustrates an example interface **350** that can facilitate search operations. The example interface **350** can, for example, be provided in response to a user entering a search query in the search bar **304** of FIG. 3A. As illustrated in FIG. 3B, a user has entered a search term “cowboys riding horses in the desert” in the search bar **354**. The example interface **350** displays videos **352a**, **352b** that satisfy the search term. Selection of the videos **352a**, **352b** can provide playback of the videos **352a**, **352b** from playback positions that satisfy the search term or provide playback of clips of the videos **352a**, **352b** that satisfy the search term. For example, in response to a user selection of video **352a**, the user can be provided with playback of video **352a** from a playback position in video **352a** that depicts cowboys riding horses in the desert. Or, the user can be provided with a clip of video **352a** that depicts cowboys riding horses in the desert.

[0046] FIG. 4 illustrates a method **400** to facilitate seek functionality in real-time streaming and playback of digital media, in accordance with one or more implementations. The operations of method **400** presented below are intended to be illustrative. In some implementations, method **400** may be accomplished with one or more additional operations not described, and/or without one or more of the operations discussed. Additionally, the order in which the operations of method **400** are illustrated in FIG. 4 and described below is not intended to be limiting.

[0047] In some implementations, method **400** may be implemented in one or more processing devices (e.g., a digital processor, an analog processor, a digital circuit designed to process

information, an analog circuit designed to process information, a state machine, and/or other mechanisms for electronically processing information). The one or more processing devices may include one or more devices executing some or all of the operations of method **400** in response to instructions stored electronically on an electronic storage medium. The one or more processing devices may include one or more devices configured through hardware, firmware, and/or software to be specifically designed for execution of one or more of the operations of method **400**.

Operations **402**, **404**, **406** may be performed by one or more hardware processors configured by machine-readable instructions including those shown in FIG. **1** and described herein.

[0048] An operation **402** can include generating an index for a digital media file encoded in a non-seekable format. In some cases, generating the index can involve, for example, performing a pass through the digital media file. File offsets of frames of the digital media file can be identified during the pass through the digital media file based on frame headers in the digital media file. In some cases, generating the index can involve storing a file offset and a time offset for each key frame of the digital media file. Each file offset provides a respective location within the digital media file for each key frame, and each time offset provides a respective time associated with each key frame.

[0049] An operation **404** can include determining a portion of the digital media file to transcode. In some cases, determining the portion of the digital media file can involve, for example, determining a file offset corresponding to a time in the digital media file indicated by a seek operation. The file offset can be determined based on an index. The digital media file can be transcoded from the file offset. In some cases, determining the portion of the digital media file can involve determining a starting file offset corresponding to a start time in the digital media file indicated by a clip operation. The starting file offset can be determined based on an index. An ending file offset corresponding to an end time in the digital media file indicated by the clip operation can be determined based on the index. The digital media file can be transcoded from the starting file offset to the ending file offset. In some cases, determining the portion of the digital media file can involve determining a file offset corresponding to a beginning of the portion of the digital media file. The file offset can be determined based on an index. The portion of the digital media file can be transcoded from the file offset.

[0050] An operation **406** can include transcoding the portion of the digital media file to a proxy format based on the index. For example, the portion of the digital media file can be transcoded to a streaming format with seek functionality provided in accordance with the features described herein.

[0051] Although the present technology has been described in detail for the purpose of illustration based on what is currently considered to be the most practical and preferred implementations, it is to be understood that such detail is solely for that purpose and that the technology is not limited to the disclosed implementations, but, on the contrary, is intended to cover modifications and equivalent arrangements that are within the spirit and scope of the appended claims. For example, it is to be understood that the present technology contemplates that, to the extent possible, one or more features of any implementation can be combined with one or more features of any other implementation.

Claims

1. A method comprising: generating, by a computer system, an index for a digital media file encoded in a non-seekable format based on extracted frame offset data associated with the digital media file; determining, by the computer system, a portion of the digital media file to transcode; and transcoding, by the computer system, the portion of the digital media file to a proxy format based on the index.
2. The method of claim 1, wherein generating, by a computer system, an index for a digital media file encoded in a non-seekable format based on extracted frame offset data associated with the

digital media file comprises: identifying frames without decoding digital media in the digital media file during a first pass.

3. The method of claim 1, wherein generating, by a computer system, an index for a digital media file encoded in a non-seekable format based on extracted frame offset data associated with the digital media file comprises: identifying during a first pass through the digital media file, file offsets of key frames of the digital media file based on frame headers.

4. The method of claim 2, further comprising: sampling, by the computing system, frames of the digital media file in a second pass through the digital media file; and extracting, by the computing system, images from the sampled frames of the digital media file.

5. The method of claim 2, wherein file offsets of key frames of the digital media file provide a number of bytes from a beginning point of the digital media file to where a respective key frame is located.

6. The method of claim 1, wherein determining the portion of the digital media file to transcode is based on a seek operation, and wherein transcoding the portion of the digital media file comprises: determining, by the computer system, a file offset corresponding to a time in the digital media file indicated by the seek operation based on the index; and transcoding, by the computer system, the digital media file from the file offset.

7. The method of claim 1, wherein determining the portion of the digital media file to transcode is based on a clip operation, and wherein transcoding the portion of the digital media file comprises: determining, by the computer system, a starting file offset corresponding to a start time in the digital media file indicated by the clip operation based on the index; determining, by the computer system, an ending file offset corresponding to an end time in the digital media file indicated by the clip operation based on the index; and transcoding, by the computer system, the digital media file from the starting file offset to the ending file offset.

8. The method of claim 1, wherein determining the portion of the digital media file to transcode is based on a playback operation, and wherein transcoding the portion of the digital media file comprises: determining, by the computer system, a file offset corresponding to a beginning of the digital media file based on the index; and transcoding, by the computer system, the digital media file from the file offset.

9. The method of claim 1, further comprising: storing, by the computer system, a transcoded portion of the digital media file; and providing, by the computer system, the transcoded portion of the digital media file in response to a user request.

10. The method of claim 1, further comprising: determining, by the computer system, a first vector embedding based on a frame in the digital media file; determining, by the computer system, a second vector embedding based on a search query, wherein determining the portion of the digital media file to transcode is based on the first vector embedding and the second vector embedding; and providing, by the computer system, the transcoded portion of the digital media file in response to the search query.

11. The method of claim 1, further comprising: determining, by the computer system, a first vector embedding based on detected scene changes in the digital media file; determining, by the computer system, a second vector embedding based on a search query, wherein determining the portion of the digital media file to transcode is based on the first vector embedding and the second vector embedding; and providing, by the computer system, the transcoded portion of the digital media file in response to the search query.

12. The method of claim 1, further comprising: extracting, by the computer system, images of the digital media file during a second pass through the digital media file based on detected scene changes; and storing, by the computer system, vector embeddings generated based on the images in the index.

13. The method of claim 12, further comprising: generating, by the computer system, a semantic vector embedding based on the extracted images of the digital media file; mapping, by the

computer system, the semantic vector embedding to a vector embedding space with vector embeddings generated based on search queries; and in response to the semantic vector embedding satisfying a threshold cosine similarity with a vector embedding of a search query, providing, by the computer system, clips or playback positions of the digital media file that correspond with images of the semantic vector embedding that satisfy the search query.

14. The method of claim 1, further comprising: receiving, by the computer system, a search query for an object; and determining, by the computer system, a playback position of the digital media file that satisfies the search query.

15. A system comprising: one or more processors; and one or more storage devices storing instructions that, when executed by the one or more processors, cause the one or more processors to: generate an index for a digital media file encoded in a non-seekable format based on extracted frame offset data associated with the digital media file; determine a portion of the digital media file to transcode; and transcode the portion of the digital media file to a proxy format based on the index.

16. The system of claim 15, wherein the instructions further cause the system to: sample frames of the digital media file in a second pass through the digital media file; and extract images from the sampled frames of the digital media file.

17. The system of claim 15, wherein the instructions to determine the portion of the digital media file to transcode is based on a seek operation, and wherein the instructions to transcode the portion of the digital media file further comprise instructions that cause the system to: determine a file offset corresponding to a time in the digital media file indicated by the seek operation based on the index; and transcode the digital media file from the file offset.

18. The system of claim 15, wherein the instructions to determine the portion of the digital media file to transcode is based on a clip operation, and wherein the instructions to transcode the portion of the digital media file further comprise instructions that cause the system to: determine a starting file offset corresponding to a start time in the digital media file indicated by the clip operation based on the index; determine an ending file offset corresponding to an end time in the digital media file indicated by the clip operation based on the index; and transcode the digital media file from the starting file offset to the ending file offset.

19. A non-transitory computer-readable storage medium including instructions that, when executed by one or more processors of a system, cause the one or more processors of the system to: generate an index for a digital media file encoded in a non-seekable format based on extracted frame offset data associated with the digital media file; determine a portion of the digital media file to transcode; and transcode the portion of the digital media file to a proxy format based on the index.

20. The non-transitory computer-readable storage medium of claim 19, wherein the instructions further cause the system to: sampling frames of the digital media file in a second pass through the digital media file; and extracting images from the sampled frames of the digital media file.
